MANAGING NON-FUNCTIONAL REQUIREMENTS IN AGILE SOFTWARE DEVELOPMENT

SWE 4803 Software Project Management

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Overview

Introduction

- **Non-functional Requirements (NFRs)** are qualities or constraints of a software system that ensure it **works well and meets certain rules.**

- In regulated environments, handling these requirements can be **tricky** when **using agile methods**.

- Right combination of **communication mechanisms** can improve agile implementation while a mismatch may be an impediment.

Introduction

The study looks at how two companies handle NFRs and communication between teams in regulated environments:

- **DevelopCo** – Makes business software for companies and governments. They turn NFRs into agile user stories (tasks in the backlog).

- **HealthCo** - Makes medical devices for healthcare. They mix waterfall (traditional) and agile methods to handle NFRs.

Related Works

- **NFR in Agile Development**
  - How NFRs are handled in agile projects and the challenges with NFR in agile
- **Inter-team Boundaries**
  - How agile teams interact with each other. Boundary spanners (e.g., product owners) bridge gaps between teams, customers, and developers
- **Regulated Environments & Agile Tailoring**
  - How agile is tailored for regulated environments using mixed agile and plan-based artefacts

Research Gap

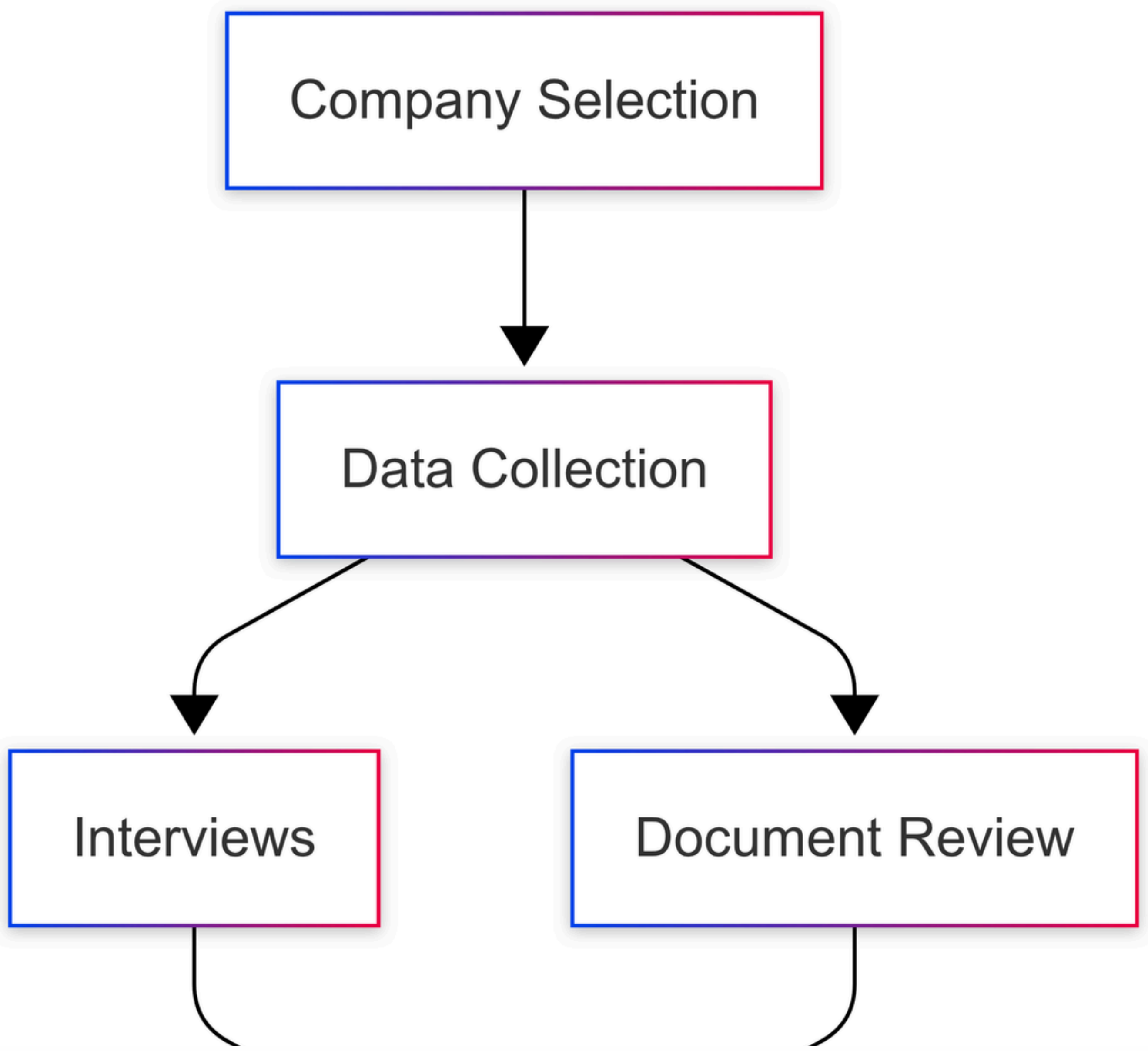
- How should **NFRs in agile** be managed when working in regulated environments?
- How to **handle inter-team communication** challenges in that context?
- How do **agile tailoring**, **NFRs**, and **cross-team collaboration** all interact?

Research Question

1. How do the practitioners in this study explain their interactions and information flows on managing NFR?

2. How do practitioners in this study describe the tailoring of agile methods to handle NFR?

Methodology



Company Selection

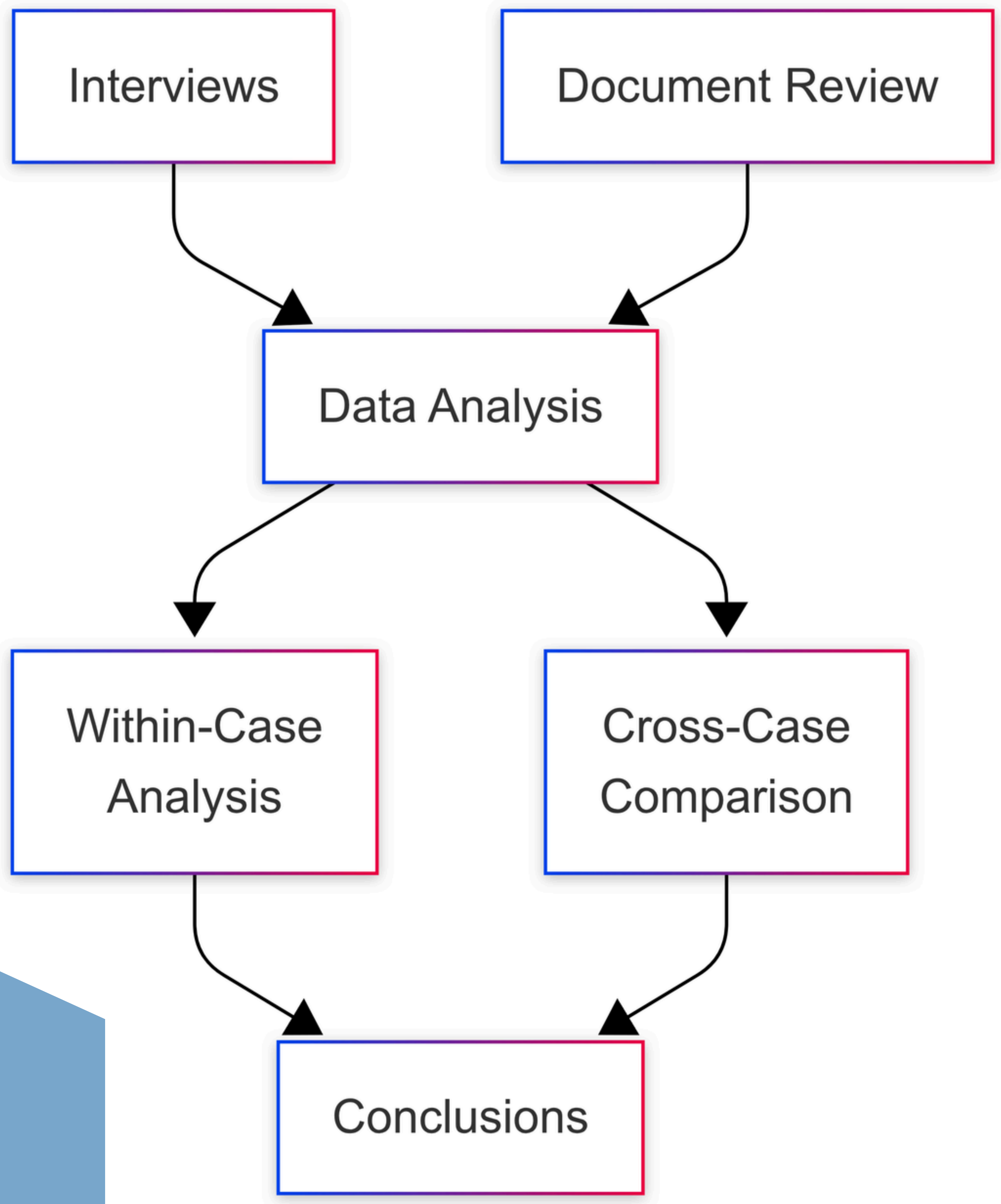
- Purposive (non-random) sampling
- 2 companies of contrastive methods: **pure agile** vs **agile with traditional** practices

Interviews

- **18** recorded interviews with semi-structured to open-ended questions
- **Manual transcription** of recordings

Document Review

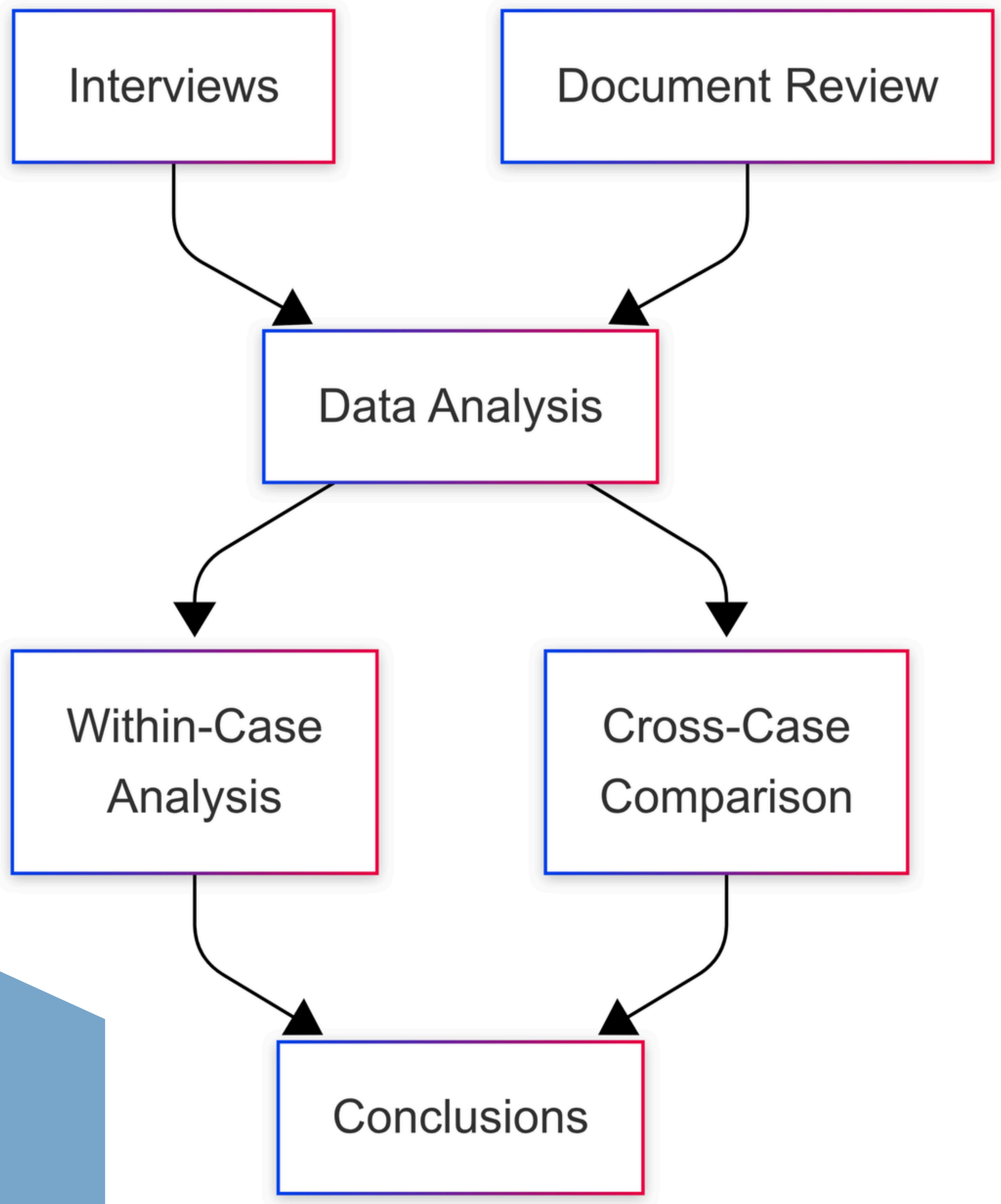
- Review of publicly available company **documentation** on development practices



Two main steps in the data analysis

Within-case Analysis

- Done separately for each company
- Coded transcripts in NVivo 11 using:
  - Line-by-line open coding
  - Constant comparison
- Developed categories until no new themes emerged



Cross-case Comparison

- Applied memo-ing approach:
  - Data reduction - summarized transcripts into **memos with key points and quotes**
  - Data display - created **swim lane diagrams** of information flows
  - Conclusion drawing - Compared diagrams to find **similarities and differences** in agile tailoring & NFR handling

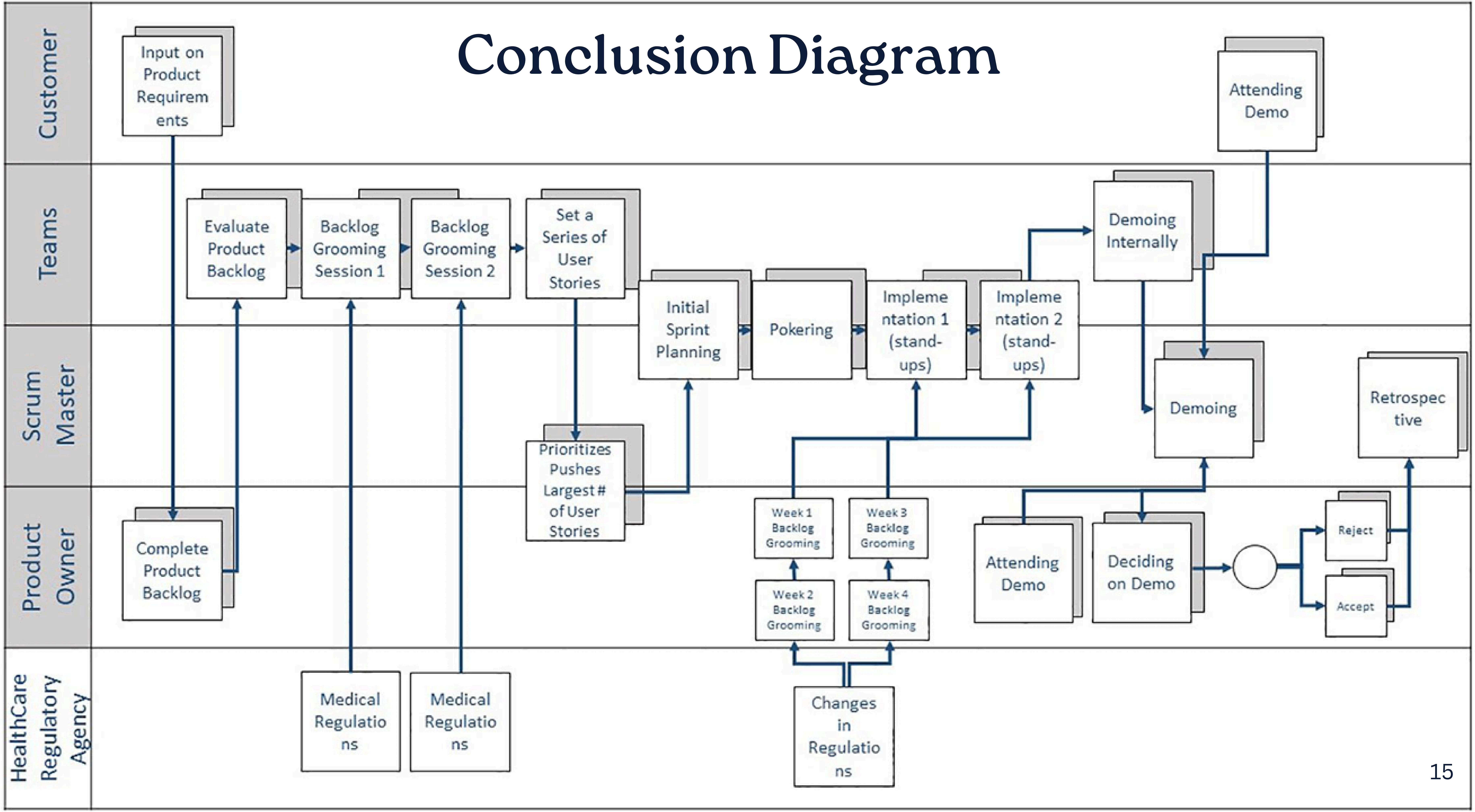
Results & Findings

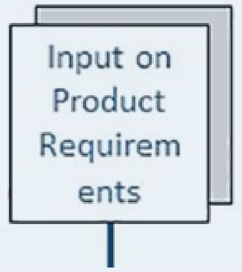
- Inter-team Dependencies
- Communication at the boundaries
- Communication and customer requirements
- Approach to NFR

	DevelopCo	HealthCo
Inter-team Dependencies	• **Delays** in design team cause idle time in development teams • **Miscommunication** cause sprint delays & idle time	• Teams are on different floors or **geographically separated** • Architect & platform team dependency → new tool within the feature phase or not?
Communication at the Boundaries	• Kanban Board (**Trello**) and **Slack** for communication • **Status assumption** by the team member • Misinterpretation of **vague customer requirements**	• **Cross-location collaboration**: structured meetings (sprints, retrospectives) & informal channels (WhatsApp) • **Physical boards** to track project progress, backlog and priority

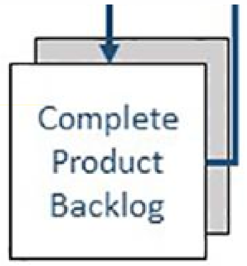
	DevelopCo	HealthCo
Communication and Customer Requirements	• Team member/ Scrum master don't ask **correct questions** • Base sprint planning on initial **customer briefing**	• **Frequent** customer meetings with Product Manager • Communication strategies followed: ◦ Stakeholder **line** meetings ◦ **Agile demos** for feedback ◦ **Retrospectives** to adapt to customer needs
Approach to NFR	• Treats all NFRs as simple **user stories** to stay agile. • Non-agile process treated as agile tasks	• Combines **Agile** (fast, flexible) with **Waterfall** (careful, documented) ways to meet strict healthcare rules • Perform documentation **regularly**

Conclusion Diagram

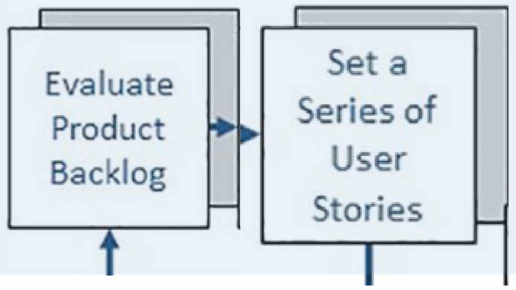




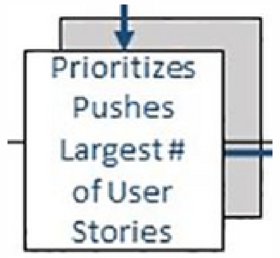
1. Customer gives their **specification**



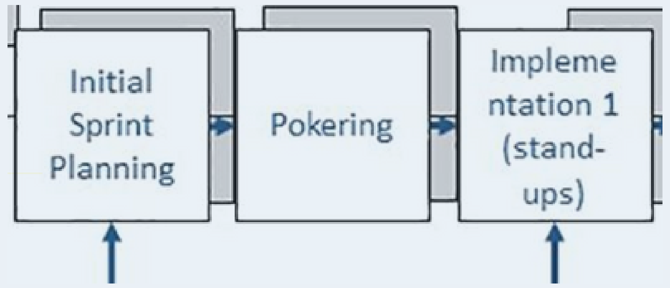
2. Product owner **creates the backlog**



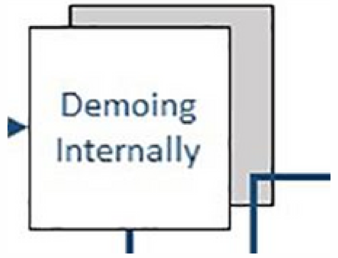
3. Teams evaluate backlog and **create user stories**



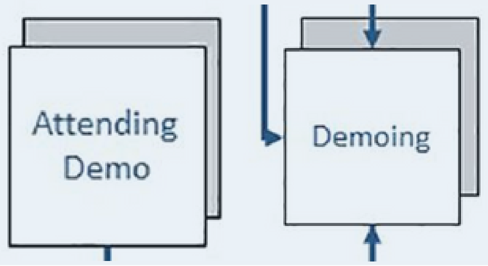
4. Product Owner and Scrum Master **prioritize completing largest number of user stories** per sprint



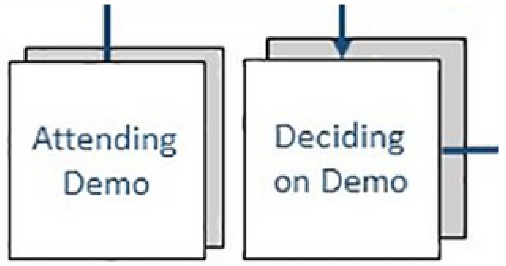
5. Team and Scrum mater start **sprint planning and implementation**



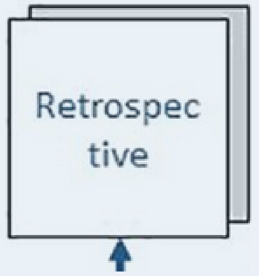
6. Teams then do **internal Demoing**



7. The scrum master holds the demo that the customer and product owner attend



8. The Product owner **accepts or rejects** the implementation



9. The teams **retrospect** based on the verdict

Proposed Solution

Authors recommend following tailoring of agile methods by adding **two proposed artefacts** to the scrum board:

Documentation Work Item: any document required to be submitted to regulatory agency

Example: **“User Experience Design is a necessary document”** added to user story with its acceptance criteria

Proposed Solution

Authors recommend following tailoring of agile methods by adding **two proposed artefacts** to the scrum board:

Safety Critical Work Item: solid safety architectures and integration of safety features

Example: "Quality assurance documentation is a requirement by the FDA" added to acceptance criteria applicable to all backlog items

Limitations

1. Depend heavily on the specific conditions, subjects, and contexts of the two studied companies, thus **limiting broader generalisation**.

2. The study follows a qualitative research method, which may not capture **statistically representative** results.

Limitations

3. Data was collected from multiple respondents but **only across two organisations**, which may not represent the full diversity of agile practices.

4. Data relies on interviews, which are subject to participants' **perceptions and possible bias** despite anonymisation.

5. Focuses specifically on **agile teams** dealing with Non-Functional Requirements (NFRs)

Conclusion

- NFR are **critical** in regulated environments and need careful management.
- HealthCo **tailors agile**, while DevelopCo handles **NFR using pure agile**.
- **Documentation and Safety Critical Work Items** introduced to handle safety critical requirements using agile.
- Recommendations for **handling NFRs** in agile processes.

Reference

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THANK YOU

Any Questions?